The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Pico-Section-Generating Substrate Femto-Section Dynamics
from Femto-Section-Generating Substrate Atto-Section Dynamics

Abstract

The preceding active-primary theorem already moved the live burden below the current pico-section package itself. That theorem derived the current pico-section package canonically from the deeper current femto-section dynamics package $\mathfrak{Z}^{\mathrm{femtosec}}$. The accompanying current femto-section frontier note then fixed the next honest move: either derive or justify $\mathfrak{Z}^{\mathrm{femtosec}}$ itself from still more primitive substrate structure, or prove a genuinely smaller theorem-level collapse strictly inside that deeper package.

The present note takes the direct derivation route. For each sector it introduces *femto-section-generating substrate atto-section dynamics*: a compact convex atto-section body over the recorded femto-section body, an affine projection to that fixed body, a nonnegative smooth atto-section functional with unique fiberwise minimizers, and an exact atto-section shell projecting diffeomorphically to the recorded exact femto-section shell. The current femto-section functional is then no longer an unexplained primitive datum. It is the effective fiberwise minimum value induced by that deeper atto-section law, and the current exact femto-section shell is the projected exact shell of the deeper package.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore narrow but benchmark facing: the current pico-section-generating substrate femto-section dynamics are no longer left as the primitive stopping point on the active line. The remaining honest burden moves one level deeper again, to the femto-section-generating substrate atto-section dynamics themselves or to a sharper collapse strictly inside that deeper package.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again reached a sharper burden than another same-output relay. The preceding active-primary theorem already showed that the current pico-section package is not primitive: it is the canonical effective output of the deeper current femto-section dynamics package $\mathfrak{Z}^{\mathrm{femtosec}}$. The accompanying current femto-section frontier note then fixed the consequence of that theorem. The next honest benchmark-facing manuscript had to either derive or justify $\mathfrak{Z}^{\mathrm{femtosec}}$ itself from still more primitive substrate structure, or prove a genuinely smaller theorem-level collapse strictly inside that package.

The present note takes the direct derivation route. Its claim is not that final substrate microphysics has been identified, and it is not that the benchmark exact-relativistic package has changed. Its claim is narrower and benchmark facing. The current femto-section package is shown to arise canonically from a more primitive atto-section layer whose projected exact-shell transport

already contains the benchmark-relevant structure of $\mathfrak{Z}^{\text{femtosec}}$. In that precise sense the current femto-section dynamics are no longer the primitive place where the active line has to stop.

Two features matter here. First, the theorem targets the current femto-section package itself rather than re-spending the already settled pico-section-package handoff, so it does not count progress merely by reproducing the same current pico-section package from yet another relay. Second, the deeper atto-section package is not merely a renaming of the current femto-section package. It adds an explicit atto-section state space, an atto-section body, an atto-to-femto-section projection, and a deeper fiberwise minimizing law whose effective energy reproduces the recorded femto-section functional on the benchmark-relevant body.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not alter the benchmark exact-closure package. It does not introduce a second operative metric or a new low-energy matter law. It only proves that the current pico-section-generating substrate femto-section dynamics can itself be generated from a more primitive atto-section-level substrate dynamics.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\text{car}, \text{cur}, \text{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\text{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\text{prim}} \equiv \mathcal{I}_{\text{car}}^{\text{for}} \times \mathcal{I}_{\text{cur}}^{\text{for}} \times \mathcal{I}_{\text{hom}}^{\text{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\text{out}} : \mathcal{S}_{\text{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\text{out}} : \mathcal{S}_{\text{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\text{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\text{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\text{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\text{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded femto-section downstream chain)**.** *Fix* $\sigma \in \{\text{car}, \text{cur}, \text{hom}\}$*. The active line already fixes the following downstream data and consequences:*

1. *the recorded pico-section target data*

$$\mathcal{Q}_\sigma^{\mathrm{picosec}}, \qquad \mathcal{B}_\sigma^{\mathrm{picosec}}, \qquad \mathscr{P}_\sigma^{\mathrm{picosec}}, \qquad \mathcal{Z}_\sigma^{\mathrm{picosec}};$$

2. *the current pico-section package $\mathfrak{Y}^{\mathrm{picosec}}$, the current observer-relevant pico-section minimizer-section core $\mathfrak{Y}^{\mathrm{pic,sec}}$, the current nano-section package $\mathfrak{X}^{\mathrm{nanosec}}$, the current micro-section package $\mathfrak{W}^{\mathrm{microsec}}$, and the current sub-section package $\mathfrak{V}^{\mathrm{subsec}}$;*

3. *the previously recorded fact that, once the current femto-section package is available, the current pico-section package, the current minimizer-section core, the current nano-section package, the current micro-section package, the current sub-section package, and the downstream shell-transport consequences used on the active line follow unchanged.*

Definition 3 (Recorded pico-section-generating substrate femto-section dynamics)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. The recorded current femto-section dynamics package $\mathfrak{Z}^{\mathrm{femtosec}}$ for sector σ consists of the following data.*

1. *A finite-dimensional Euclidean femto-section state space $\mathcal{R}_\sigma^{\mathrm{femtosec}}$, a compact convex femto-section body*

$$\mathcal{B}_\sigma^{\mathrm{femtosec}} \subset \mathcal{R}_\sigma^{\mathrm{femtosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{femtosec}} : \mathcal{R}_\sigma^{\mathrm{femtosec}} \to \mathcal{Q}_\sigma^{\mathrm{picosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{femtosec}}(\mathcal{B}_\sigma^{\mathrm{femtosec}}) = \mathcal{B}_\sigma^{\mathrm{picosec}}.$$

For each $r \in \mathcal{B}_\sigma^{\mathrm{picosec}}$, write

$$\mathcal{F}_\sigma^{\mathrm{femtosec}}(r) \equiv \left(\Pi_\sigma^{\mathrm{femtosec}}\right)^{-1}(r) \cap \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth femto-section functional*

$$\mathscr{F}_\sigma^{\mathrm{femtosec}} : \mathcal{R}_\sigma^{\mathrm{femtosec}} \to [0, \infty)$$

such that, for every $r \in \mathcal{B}_\sigma^{\mathrm{picosec}}$, the restriction of $\mathscr{F}_\sigma^{\mathrm{femtosec}}$ to $\mathcal{F}_\sigma^{\mathrm{femtosec}}(r)$ is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on r; write it as

$$\mathfrak{f}_\sigma^{\mathrm{femtosec}} : \mathcal{B}_\sigma^{\mathrm{picosec}} \to \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

These data satisfy

$$\Pi_\sigma^{\mathrm{femtosec}} \circ \mathfrak{f}_\sigma^{\mathrm{femtosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{picosec}}},$$

and the effective minimum value recovers the recorded pico-section functional:

$$\mathscr{P}_\sigma^{\mathrm{picosec}}(r) = \mathscr{F}_\sigma^{\mathrm{femtosec}}(\mathfrak{f}_\sigma^{\mathrm{femtosec}}(r)) = \min_{t \in \mathcal{F}_\sigma^{\mathrm{femtosec}}(r)} \mathscr{F}_\sigma^{\mathrm{femtosec}}(t) \qquad \textit{for every } r \in \mathcal{B}_\sigma^{\mathrm{picosec}}.$$

3. *A closed embedded exact femto-section shell*

$$\mathcal{Z}_\sigma^{\mathrm{femtosec}} \subset \mathcal{B}_\sigma^{\mathrm{femtosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{femtosec}} = \left\{ t \in \mathcal{B}_\sigma^{\mathrm{femtosec}} : \begin{array}{l} \Pi_\sigma^{\mathrm{femtosec}}(t) \in \mathcal{Z}_\sigma^{\mathrm{picosec}}, \\ \mathscr{F}_\sigma^{\mathrm{femtosec}}(t) = \min\limits_{u \in \mathcal{F}_\sigma^{\mathrm{femtosec}}(\Pi_\sigma^{\mathrm{femtosec}}(t))} \mathscr{F}_\sigma^{\mathrm{femtosec}}(u) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{femtosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{femtosec}}} : \mathcal{Z}_\sigma^{\mathrm{femtosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{picosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as a deeper substrate-side source for the current pico-section package rather than as a replacement benchmark theory.*

Remark 1. *Definitions 1–3 are fixed inputs. The one operative metric, the ordinary matter route, the recorded femto-section body* $\mathcal{B}_\sigma^{\mathrm{femtosec}}$*, the femto-section functional* $\mathscr{F}_\sigma^{\mathrm{femtosec}}$*, the exact femto-section shell* $\mathcal{Z}_\sigma^{\mathrm{femtosec}}$*, and the already recorded downstream consequences are not reopened here. The present note asks only whether the current femto-section package can itself be generated from still more primitive substrate structure.*

3. Femto-Section-Generating Substrate Atto-Section Dynamics

Definition 4 (Femto-section-generating substrate atto-section dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* femto-section-generating substrate atto-section dynamics package $\mathfrak{A}^{\mathrm{attosec}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean atto-section state space* $\mathcal{A}_\sigma^{\mathrm{attosec}}$*, a compact convex atto-section body*

$$\mathcal{B}_\sigma^{\mathrm{attosec}} \subset \mathcal{A}_\sigma^{\mathrm{attosec}}$$

with nonempty interior, and an affine surjection

$$\Pi_\sigma^{\mathrm{attosec}} : \mathcal{A}_\sigma^{\mathrm{attosec}} \to \mathcal{R}_\sigma^{\mathrm{femtosec}}$$

satisfying

$$\Pi_\sigma^{\mathrm{attosec}}(\mathcal{B}_\sigma^{\mathrm{attosec}}) = \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

For each $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$*, write*

$$\mathcal{F}_\sigma^{\mathrm{attosec}}(t) \equiv (\Pi_\sigma^{\mathrm{attosec}})^{-1}(t) \cap \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

Each such fiber is required to be nonempty, compact, and convex.

2. *A nonnegative smooth atto-section functional*

$$\mathscr{A}_\sigma^{\mathrm{attosec}} : \mathcal{A}_\sigma^{\mathrm{attosec}} \to [0, \infty)$$

such that, for every $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$, the restriction of $\mathscr{A}_\sigma^{\mathrm{attosec}}$ to $\mathcal{F}_\sigma^{\mathrm{attosec}}(t)$ is strictly convex and attains a unique minimizer. The resulting minimizer depends smoothly on t; write it as

$$\mathfrak{a}_\sigma^{\mathrm{attosec}} : \mathcal{B}_\sigma^{\mathrm{femtosec}} \to \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

These data satisfy

$$\Pi_\sigma^{\mathrm{attosec}} \circ \mathfrak{a}_\sigma^{\mathrm{attosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{femtosec}}},$$

and the effective minimum value recovers the recorded femto-section functional:

$$\mathscr{F}_\sigma^{\mathrm{femtosec}}(t) = \mathscr{A}_\sigma^{\mathrm{attosec}}(\mathfrak{a}_\sigma^{\mathrm{attosec}}(t)) = \min_{v \in \mathcal{F}_\sigma^{\mathrm{attosec}}(t)} \mathscr{A}_\sigma^{\mathrm{attosec}}(v) \qquad \text{for every } t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}.$$

3. *A closed embedded exact atto-section shell*

$$\mathcal{Z}_\sigma^{\mathrm{attosec}} \subset \mathcal{B}_\sigma^{\mathrm{attosec}}$$

satisfying

$$\mathcal{Z}_\sigma^{\mathrm{attosec}} = \left\{ v \in \mathcal{B}_\sigma^{\mathrm{attosec}} : \begin{array}{l} \Pi_\sigma^{\mathrm{attosec}}(v) \in \mathcal{Z}_\sigma^{\mathrm{femtosec}}, \\ \mathscr{A}_\sigma^{\mathrm{attosec}}(v) = \min\limits_{w \in \mathcal{F}_\sigma^{\mathrm{attosec}}(\Pi_\sigma^{\mathrm{attosec}}(v))} \mathscr{A}_\sigma^{\mathrm{attosec}}(w) \end{array} \right\},$$

and

$$\Pi_\sigma^{\mathrm{attosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{attosec}}} : \mathcal{Z}_\sigma^{\mathrm{attosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{femtosec}}$$

is a diffeomorphism.

4. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and presentation as a deeper substrate-side source for the current femto-section package rather than as a replacement benchmark theory.*

Remark 2. *Definition 4 is intentionally written over the already fixed femto-section body, femto-section functional, and exact femto-section shell. It does not reopen the already-generated current pico-section package or its downstream minimizer-section-core presentation as the live burden. Its positive role is narrower: the current femto-section package is generated as the unique fiberwise effective output of a deeper atto-section selector law.*

4. Canonical Recovery of the Current Femto-Section Package

Proposition 1 (The canonical atto minimizer section is well defined)**.** *For every benchmark-faithful atto-section dynamics package, the map $\mathfrak{a}_\sigma^{\mathrm{attosec}}$ of Definition 4 is a smooth section of $\Pi_\sigma^{\mathrm{attosec}}$, so that*

$$\Pi_\sigma^{\mathrm{attosec}} \circ \mathfrak{a}_\sigma^{\mathrm{attosec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{femtosec}}}.$$

Proof. Definition 4(1) makes each fiber $\mathcal{F}_\sigma^{\mathrm{attosec}}(t)$ nonempty, compact, and convex. Definition 4(2) states that $\mathscr{A}_\sigma^{\mathrm{attosec}}$ has a unique minimizer on each such fiber and that the minimizer depends smoothly on t. Definition 4(2) therefore gives a well-defined smooth map

$$\mathfrak{a}_\sigma^{\mathrm{attosec}} : \mathcal{B}_\sigma^{\mathrm{femtosec}} \to \mathcal{B}_\sigma^{\mathrm{attosec}}.$$

Because every minimizer lies in the fiber over the point at which it is selected, we have

$$\Pi_\sigma^{\mathrm{attosec}}\big(\mathfrak{a}_\sigma^{\mathrm{attosec}}(t)\big) = t$$

for every $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$, which is the displayed identity. □

Proposition 2 (The exact femto-section shell is the projected exact atto-section shell)**.** *For every benchmark-faithful atto-section dynamics package,*

$$\Pi_\sigma^{\mathrm{attosec}}\big(\mathcal{Z}_\sigma^{\mathrm{attosec}}\big) = \mathcal{Z}_\sigma^{\mathrm{femtosec}},$$

and the shell restriction

$$\Pi_\sigma^{\mathrm{attosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{attosec}}} : \mathcal{Z}_\sigma^{\mathrm{attosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{femtosec}}$$

is a diffeomorphism with inverse

$$\mathfrak{a}_\sigma^{\mathrm{attosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{femtosec}}} : \mathcal{Z}_\sigma^{\mathrm{femtosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{attosec}}.$$

Proof. Definition 4(3) states directly that

$$\Pi_\sigma^{\mathrm{attosec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{attosec}}} : \mathcal{Z}_\sigma^{\mathrm{attosec}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{femtosec}}$$

is a diffeomorphism, so the image identity follows immediately. To identify the inverse, let $t \in \mathcal{Z}_\sigma^{\mathrm{femtosec}}$. By Proposition 1, $\mathfrak{a}_\sigma^{\mathrm{attosec}}(t)$ lies in the fiber over t. Since $t \in \mathcal{Z}_\sigma^{\mathrm{femtosec}}$, Definition 4(3) shows that this point belongs to $\mathcal{Z}_\sigma^{\mathrm{attosec}}$. Hence the shell restriction of $\mathfrak{a}_\sigma^{\mathrm{attosec}}$ is indeed the inverse diffeomorphism. □

Proposition 3 (The induced femto-section package is canonical)**.** *Fix a benchmark-faithful atto-section dynamics package. If*

$$s : \mathcal{B}_\sigma^{\mathrm{femtosec}} \to \mathcal{B}_\sigma^{\mathrm{attosec}}$$

is any smooth section satisfying

$$\Pi_\sigma^{\mathrm{attosec}} \circ s = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{femtosec}}}$$

and

$$\mathscr{A}_\sigma^{\mathrm{attosec}}\big(s(t)\big) = \min_{v \in \mathcal{F}_\sigma^{\mathrm{attosec}}(t)} \mathscr{A}_\sigma^{\mathrm{attosec}}(v) \qquad \textit{for every } t \in \mathcal{B}_\sigma^{\mathrm{femtosec}},$$

then $s = \mathfrak{a}_\sigma^{\mathrm{attosec}}$. In particular, the induced current femto-section package carries no residual atto-fiber minimizing-choice freedom once the deeper atto-section dynamics are fixed.

Proof. For each $t \in \mathcal{B}_\sigma^{\mathrm{femtosec}}$, the point $s(t)$ belongs to $\mathcal{F}_\sigma^{\mathrm{attosec}}(t)$ and minimizes $\mathscr{A}_\sigma^{\mathrm{attosec}}$ there. By Definition 4(2), that minimizer is unique. Therefore

$$s(t) = \mathfrak{a}_\sigma^{\mathrm{attosec}}(t)$$

for every t, hence $s = \mathfrak{a}_\sigma^{\mathrm{attosec}}$. □

Proposition 4 (All current downstream uses factor through the induced femto-section package)**.** *For every benchmark-faithful atto-section dynamics package, all current benchmark-facing downstream uses of the current femto-section package $\mathfrak{Z}^{\mathrm{femtosec}}$ on the active line factor through the induced*

femto-section data together with the fixed atto-section package. In particular, the already recorded recovery of the current pico-section package $\mathfrak{Y}^{\mathrm{picosec}}$, the current minimizer-section core $\mathfrak{Y}^{\mathrm{pic,sec}}$, the current nano-section package $\mathfrak{X}^{\mathrm{nanosec}}$, the current micro-section package $\mathfrak{W}^{\mathrm{microsec}}$, the current sub-section package $\mathfrak{V}^{\mathrm{subsec}}$, and the previously recorded shell-transport consequences does not require any additional primitive femto-section datum once the induced femto-section package is fixed.

Proof. By Definition 4(1)–(3), the deeper atto-section package already supplies the recorded femto-section body, femto-section functional, and exact femto-section shell. Proposition 1 gives the canonical atto minimizer section over that body, and Proposition 2 gives the exact shell projection and inverse lift. Definition 2(3) then records that, once the current femto-section package is available, the current pico-section package, the current minimizer-section core, the current nano-section package, the current micro-section package, the current sub-section package, and the downstream shell-transport consequences used on the active line follow unchanged. Therefore all current benchmark-facing downstream uses at femto-section scope already factor through the induced femto-section package together with the fixed atto-section data, and no additional primitive femto-section datum is required. □

Proposition 5 (The benchmark package remains fixed)**.** *Every benchmark-faithful femto-section-generating substrate atto-section dynamics package preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 4(4). The present note only derives the current femto-section package from deeper substrate selector data. It does not alter the benchmark exact-closure package. □

5. Main Theorem

Theorem 1 (Pico-section-generating substrate femto-section dynamics from femto-section-generating substrate atto-section dynamics)**.** *Fix the primitive continuation data of Definition 1, the recorded femto-section downstream chain of Definition 2, the recorded current femto-section package of Definition 3, and a benchmark-faithful femto-section-generating substrate atto-section dynamics package in the sense of Definition 4. Then, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the current pico-section-generating substrate femto-section dynamics are no longer an unexplained primitive stopping point on the active line: they are recovered canonically from the deeper atto-section dynamics. More precisely:*

1. *the deeper atto-section dynamics generate the current femto-section body, the canonical atto minimizer lift over that body, the effective femto-section functional, and the exact femto-section shell by Definition 4, Proposition 1, and Proposition 2;*

2. *the induced current femto-section package is canonical and carries no residual atto-fiber minimizing-choice freedom once the deeper package is fixed by Proposition 3;*

3. *all current benchmark-facing downstream uses of the current femto-section package already factor through the induced datum by Proposition 4;*

4. *the benchmark boundary remains fixed by Proposition 5;*

5. *consequently the remaining honest burden below the current femto-section-package frontier is no longer the current pico-section-generating substrate femto-section dynamics* $\mathfrak{Z}^{\mathrm{femtosec}}$ *as such. What remains is to derive or justify the deeper femto-section-generating substrate atto-section dynamics* $\mathfrak{A}^{\mathrm{attosec}}$ *from still more primitive substrate structure, or else prove a still sharper theorem-level collapse strictly inside that deeper package.*

Proof. Item (1) is the content of Definition 4 together with Propositions 1 and 2. Item (2) is Proposition 3. Item (3) is Proposition 4. Item (4) is Proposition 5. Item (5) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the femto-section-generating substrate atto-section dynamics* $\mathfrak{A}^{\mathrm{attosec}}$ *from still more primitive substrate structure in a way that changes benchmark-facing status;*

2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{A}^{\mathrm{attosec}}$ *and prove a sharper collapse to it;*

3. *do either of the previous two while preserving the same benchmark one-metric package and the same claim boundary.*

Another same-output deeper-origin relay that merely preserves the current atto-section package $\mathfrak{A}^{\mathrm{attosec}}$ *and therefore merely reproduces the same current femto-section package* $\mathfrak{Z}^{\mathrm{femtosec}}$*, the same current pico-section package* $\mathfrak{Y}^{\mathrm{picosec}}$*, the same current minimizer-section core* $\mathfrak{Y}^{\mathrm{pic,sec}}$*, the same current nano-section package* $\mathfrak{X}^{\mathrm{nanosec}}$*, and the same downstream micro-section and sub-section packages, another re-expansion back to the larger minimizer-image-core or full pico-section presentations as if they were still independently live, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden. If a quantum continuation is pursued, it matters benchmark-facingly only insofar as it derives this same deeper atto-section dynamics, collapses them further, or closes a benchmark derivation gate in a genuinely new way.*

Proof. Theorem 1 shows that the current femto-section package is no longer the deepest unexplained datum on the active line. Therefore a subsequent manuscript must improve on the deeper atto-section dynamics rather than merely restate the already generated femto-section package. □

6. Conclusion

The positive result of this note is that the current pico-section-generating substrate femto-section dynamics are no longer primitive on the active derivational line. Once one works with a femto-section-generating substrate atto-section dynamics package over the already fixed femto-section body, the current femto-section functional and exact femto-section shell are forced as the effective fiberwise output of a deeper substrate selection law. All current downstream benchmark-facing uses already factor through that induced femto-section package together with the fixed atto-section package.

This preserves the same benchmark one-metric package and the same claim boundary. The result does not yet provide a completed first-principles derivation of GR from substrate microphysics, and it does not license stronger promotion language. Its narrower benchmark-facing gain is that the remaining honest burden now sits one level lower again: not on the current femto-section package itself, but on the deeper atto-section dynamics that canonically generate it, or on a still smaller collapse strictly inside that deeper package.